IMovies – Movie Ticket Booking System

Project Development Phase

Implementation notes based on the supplied source

Project documentation based on the uploaded bookMyShow-main source ZIP

Prepared: 31 October 2026

Project Development Phase

5.1 Technology stack

Frontend: React 18, React Router, Redux Toolkit, Axios, Ant Design and Moment. Backend: Node.js and Express. Database: MongoDB/Mongoose. Security libraries: bcryptjs and jsonwebtoken. Stripe SDK is listed as a dependency.

5.2 Frontend structure

The client contains pages for Home, Login, Register, movie theatres, BookShow, Profile, Admin, movie forms, theatre lists and bookings. API helper modules are separated for users, movies, theatres and bookings. Redux slices store user and loader state.

5.3 Backend structure

server/index.js loads dotenv and database configuration, configures CORS and JSON parsing, mounts four route groups, and listens on port 8082. Routes separate user, movie, theatre and booking operations.

5.4 Database models

Mongoose schemas define users, movies, theatres, shows and bookings. ObjectId references connect users to bookings, shows to movies/theatres, and bookings to shows/users.

5.5 Authentication implementation

Registration hashes the supplied password using bcryptjs. Login compares the password hash and signs a JWT using an environment secret. Protected routes use authMiddleware. Never commit secrets or real user credentials.

5.6 Booking implementation

The booking route reads the show, combines selected seats with the existing bookedSeats array, updates the show and creates a booking document. Inspect the exact route and test duplicate-seat and concurrent-booking behavior; a simple read-then-write sequence can be vulnerable to races.

5.7 Local configuration

The README specifies MONGO_URL, JWT_SECRET and PORT=8082. Check config/dbConfig.js and auth middleware for exact environment variable spelling before creating .env. The current server/index.js CORS origin is the deployed frontend; add http://localhost:3000 for local development.

5.8 Running locally

- Install a Node.js LTS release compatible with the project.
- Create server/.env with MONGO_URL, JWT_SECRET and any required Stripe key only if payment code requires it.
- In terminal 1: cd server, npm install, npm start.
- In terminal 2: cd client, npm install, npm start.
- Open http://localhost:3000 and keep both terminals running.
- If the client API helper points to a hosted URL, change it to http://localhost:8082 for a fully local demo.

5.9 Known implementation checks

The client package uses react-scripts, not Vite. The README mentions fixed 10×10 seats and up to six seats. The server has no meaningful test script configured. Confirm seed data, admin credentials, payment setup, route protection, error handling and booking concurrency before presenting them as completed.

5.10 Suggested screenshots

Capture: home/movie list; registration/login; movie details; theatre/show selection; seat grid; booking confirmation; booking history; admin movie list/form; admin theatre/show management. Use genuine screenshots from your own running instance.

Accuracy note: This document describes the code and README found in the supplied ZIP. Items marked as recommendations or proposed tests are not claims that they have already been implemented or passed. Confirm all workflows on your laptop before submission.